Run report: S_3 Coppersmith θ_2 falsifier, run-20261023T2122Z

Descriptive record of one run under the frozen protocol `../../PROTOCOL.md` (committed as `c785a0e23` before this run). Observations only; the reading against the decision rule is in `../../REPORT.md`.

item	value
Instrument	`examples/coppersmith_s3_theta2_falsifier.rs` at commit `c785a0e232f340cd3b32b1a88ebd3926e97e93ed` (`cargo build --release`)
Command	`./target/release/examples/coppersmith_s3_theta2_falsifier --bits 24,32,40,56 --ell 1,2,3 --beta 0.04:0.34:0.02 --trials 10 --neg-trials 10 --neg-beta 0.30 --seed 20261023 --out reports/run-20261023T2122Z`
Host	4 vCPU, x86_64, Intel(R) Xeon(R) Processor @ 2.10GHz; cloud container, not isolated; one thread (`host.txt`, `load_before.txt`, `load_after.txt`)
Wall time	15m20.629s (`time.txt`); the instrument's own clock: 920.624825337 s
Primes	24-bit 16777213, 32-bit 4294967291, 40-bit 1099511627689, 56-bit 72057594037927931 (largest prime below 2^`bits`); one random curve per size (a, b in `trials.jsonl`)
Arms	planted: 4 sizes × $\ell \in \{1, 2, 3\}$ × 16 values of β × N = 10 (1920 trials); negative control: 4 × 3 × 10 at β = 0.30 (120 trials)
Seeds	`seed = 20261023`; per-trial seeds recorded in `trials.jsonl`
Raw rows	`trials.jsonl` (2040 rows, one per trial); cell summaries `run.json`; the table below is `stdout.md`
Sieve primes	2305843009213693951, 2305843009213693921, 2305843009213693907 (`run.json`)
Failures	none; every trial completed; exit 0

Columns. `recovered/N`: the pipeline's reported set of genuine pairs contains the planted $\{x(P_1), x(P_2)\}$ (decision metric). `recovered (HG rows)`: same, restricted to row pairs satisfying Howgrave-Graham. `HG first two`: the first two reduced rows satisfy Howgrave-Graham (a priori, root not used). `≥2 rows vanish`: at least two reduced rows vanish at the planted root over Z (diagnostic, uses the root). Times are means over the cell's trials, wall-clock milliseconds on this host, not a benchmark.

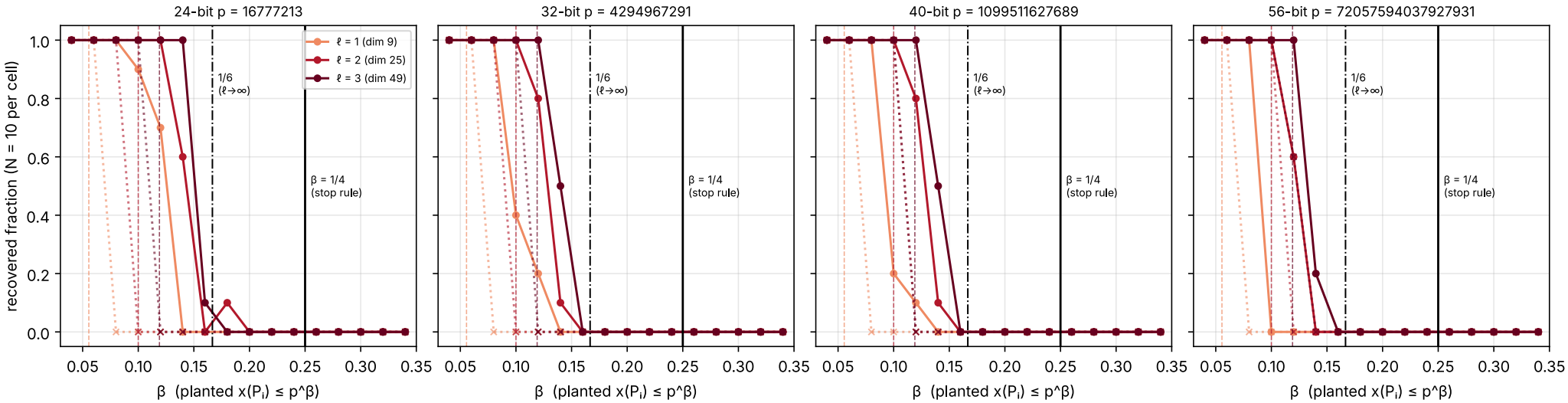


Figure 1: Recovered fraction per cell; see run.md

Recorded cells

bits	p	ℓ	dim	β	B	recovered/ N	recovered (HG rows)/N	HG first two/N	≥2 rows vanish/N	mean LLL ms	mean extract ms
24	16777213	1	9	0.04	1	10/10	10/10	10/10	10/10	4.4	1.8
24	16777213	1	9	0.06	2	10/10	10/10	10/10	10/10	4.5	3.7
24	16777213	1	9	0.08	3	10/10	0/10	0/10	10/10	2.4	1.7
24	16777213	1	9	0.10	5	9/10	0/10	0/10	9/10	1.6	1.3
24	16777213	1	9	0.12	7	7/10	0/10	0/10	8/10	1.6	1.3
24	16777213	1	9	0.14	10	0/10	0/10	0/10	1/10	1.1	0.5
24	16777213	1	9	0.16	14	0/10	0/10	0/10	0/10	0.9	0.6
24	16777213	1	9	0.18	19	0/10	0/10	0/10	1/10	0.9	0.6
24	16777213	1	9	0.20	27	0/10	0/10	0/10	0/10	0.9	0.5
24	16777213	1	9	0.22	38	0/10	0/10	0/10	0/10	0.8	0.6
24	16777213	1	9	0.24	54	0/10	0/10	0/10	0/10	0.7	0.6
24	16777213	1	9	0.26	75	0/10	0/10	0/10	0/10	1.7	0.6
24	16777213	1	9	0.28	105	0/10	0/10	0/10	0/10	1.2	0.7
24	16777213	1	9	0.30	147	0/10	0/10	0/10	0/10	0.7	0.7
24	16777213	1	9	0.32	205	0/10	0/10	0/10	0/10	0.6	0.5
24	16777213	1	9	0.34	286	0/10	0/10	0/10	0/10	0.4	0.4
24	16777213	1	9	0.30 (negative control)	147	genuine 0/10 (raw candidates in 10)	–	–	–	0.7	0.7
24	16777213	2	25	0.04	1	10/10	10/10	10/10	10/10	271.7	16.6
24	16777213	2	25	0.06	2	10/10	10/10	10/10	10/10	175.3	18.5
24	16777213	2	25	0.08	3	10/10	10/10	10/10	10/10	151.9	20.7
24	16777213	2	25	0.10	5	10/10	0/10	0/10	10/10	92.5	15.9
24	16777213	2	25	0.12	7	10/10	0/10	0/10	10/10	64.7	10.6
24	16777213	2	25	0.14	10	6/10	0/10	0/10	7/10	42.2	10.4
24	16777213	2	25	0.16	14	0/10	0/10	0/10	2/10	23.6	1.4
24	16777213	2	25	0.18	19	1/10	0/10	0/10	1/10	20.9	0.7
24	16777213	2	25	0.20	27	0/10	0/10	0/10	0/10	20.3	0.7
24	16777213	2	25	0.22	38	0/10	0/10	0/10	0/10	14.3	0.7
24	16777213	2	25	0.24	54	0/10	0/10	0/10	0/10	18.4	0.8
24	16777213	2	25	0.26	75	0/10	0/10	0/10	0/10	13.8	0.7
24	16777213	2	25	0.28	105	0/10	0/10	0/10	0/10	11.5	0.7
24	16777213	2	25	0.30	147	0/10	0/10	0/10	0/10	10.8	0.8
24	16777213	2	25	0.32	205	0/10	0/10	0/10	0/10	9.4	0.5
24	16777213	2	25	0.34	286	0/10	0/10	0/10	0/10	7.2	0.4
24	16777213	2	25	0.30 (negative control)	147	genuine 0/10 (raw candidates in 10)	–	–	–	11.6	0.8
24	16777213	3	49	0.04	1	10/10	10/10	10/10	10/10	5382.0	90.9
24	16777213	3	49	0.06	2	10/10	10/10	10/10	10/10	2893.0	87.1

24	16777213	3	49	0.08	3	10/10	10/10	10/10	10/10	2348.8	88.8
24	16777213	3	49	0.10	5	10/10	10/10	10/10	10/10	1670.7	97.2
24	16777213	3	49	0.12	7	10/10	0/10	0/10	10/10	1088.3	88.3
24	16777213	3	49	0.14	10	10/10	0/10	0/10	10/10	641.3	61.2
24	16777213	3	49	0.16	14	1/10	0/10	0/10	6/10	233.8	1.5
24	16777213	3	49	0.18	19	0/10	0/10	0/10	3/10	192.6	1.1
24	16777213	3	49	0.20	27	0/10	0/10	0/10	2/10	145.4	1.0
24	16777213	3	49	0.22	38	0/10	0/10	0/10	0/10	139.0	1.1
24	16777213	3	49	0.24	54	0/10	0/10	0/10	0/10	137.2	1.2
24	16777213	3	49	0.26	75	0/10	0/10	0/10	0/10	125.1	1.2
24	16777213	3	49	0.28	105	0/10	0/10	0/10	0/10	113.5	1.1
24	16777213	3	49	0.30	147	0/10	0/10	0/10	0/10	100.2	1.1
24	16777213	3	49	0.32	205	0/10	0/10	0/10	0/10	96.1	1.2
24	16777213	3	49	0.34	286	0/10	0/10	0/10	0/10	90.0	1.0
24	16777213	3	49	0.30 (negative control)	147	genuine 0/10 (raw candidates in 10)	–	–	–	105.8	1.2
32	4294967291	1	9	0.04	2	10/10	10/10	10/10	10/10	5.3	2.2
32	4294967291	1	9	0.06	3	10/10	10/10	10/10	10/10	3.9	2.0
32	4294967291	1	9	0.08	5	10/10	0/10	0/10	10/10	2.7	1.8
32	4294967291	1	9	0.10	9	4/10	0/10	0/10	5/10	2.3	1.2
32	4294967291	1	9	0.12	14	2/10	0/10	0/10	2/10	1.6	1.2
32	4294967291	1	9	0.14	22	0/10	0/10	0/10	0/10	1.1	0.8
32	4294967291	1	9	0.16	34	0/10	0/10	0/10	0/10	1.3	0.5
32	4294967291	1	9	0.18	54	0/10	0/10	0/10	0/10	0.9	0.5
32	4294967291	1	9	0.20	84	0/10	0/10	0/10	0/10	0.9	0.6
32	4294967291	1	9	0.22	131	0/10	0/10	0/10	0/10	1.0	0.6
32	4294967291	1	9	0.24	205	0/10	0/10	0/10	0/10	0.9	0.6
32	4294967291	1	9	0.26	319	0/10	0/10	0/10	0/10	0.7	0.6
32	4294967291	1	9	0.28	497	0/10	0/10	0/10	0/10	0.6	0.7
32	4294967291	1	9	0.30	776	0/10	0/10	0/10	0/10	0.5	0.7
32	4294967291	1	9	0.32	1209	0/10	0/10	0/10	0/10	0.4	0.8
32	4294967291	1	9	0.34	1884	0/10	0/10	0/10	0/10	0.3	0.6
32	4294967291	1	9	0.30 (negative control)	776	genuine 0/10 (raw candidates in 10)	–	–	–	0.7	0.8
32	4294967291	2	25	0.04	2	10/10	10/10	10/10	10/10	235.2	16.3
32	4294967291	2	25	0.06	3	10/10	10/10	10/10	10/10	225.9	16.9
32	4294967291	2	25	0.08	5	10/10	10/10	10/10	10/10	188.0	17.6
32	4294967291	2	25	0.10	9	10/10	0/10	0/10	10/10	140.9	17.5
32	4294967291	2	25	0.12	14	8/10	0/10	0/10	10/10	90.9	11.2
32	4294967291	2	25	0.14	22	1/10	0/10	0/10	4/10	62.5	11.5
32	4294967291	2	25	0.16	34	0/10	0/10	0/10	1/10	33.3	1.2

32	4294967291	2	25	0.18	54	0/10	0/10	0/10	0/10	25.3	0.7
32	4294967291	2	25	0.20	84	0/10	0/10	0/10	0/10	21.9	0.8
32	4294967291	2	25	0.22	131	0/10	0/10	0/10	0/10	19.3	0.7
32	4294967291	2	25	0.24	205	0/10	0/10	0/10	0/10	16.2	0.8
32	4294967291	2	25	0.26	319	0/10	0/10	0/10	0/10	17.9	0.9
32	4294967291	2	25	0.28	497	0/10	0/10	0/10	0/10	13.5	0.8
32	4294967291	2	25	0.30	776	0/10	0/10	0/10	0/10	13.7	1.0
32	4294967291	2	25	0.32	1209	0/10	0/10	0/10	0/10	10.3	0.9
32	4294967291	2	25	0.34	1884	0/10	0/10	0/10	0/10	10.4	0.9
32	4294967291	2	25	0.30 (negative control)	776	genuine 0/10 (raw candidates in 10)	–	–	–	13.6	0.9
32	4294967291	3	49	0.04	2	10/10	10/10	10/10	10/10	4627.3	88.7
32	4294967291	3	49	0.06	3	10/10	10/10	10/10	10/10	3635.4	89.1
32	4294967291	3	49	0.08	5	10/10	10/10	10/10	10/10	2976.1	87.8
32	4294967291	3	49	0.10	9	10/10	10/10	10/10	10/10	2100.9	87.1
32	4294967291	3	49	0.12	14	10/10	0/10	0/10	10/10	1494.6	100.9
32	4294967291	3	49	0.14	22	5/10	0/10	0/10	9/10	849.4	61.3
32	4294967291	3	49	0.16	34	0/10	0/10	0/10	3/10	335.5	2.0
32	4294967291	3	49	0.18	54	0/10	0/10	0/10	0/10	225.6	1.1
32	4294967291	3	49	0.20	84	0/10	0/10	0/10	0/10	189.4	1.2
32	4294967291	3	49	0.22	131	0/10	0/10	0/10	0/10	182.8	1.3
32	4294967291	3	49	0.24	205	0/10	0/10	0/10	0/10	158.5	1.3
32	4294967291	3	49	0.26	319	0/10	0/10	0/10	0/10	134.8	1.2
32	4294967291	3	49	0.28	497	0/10	0/10	0/10	0/10	127.1	1.3
32	4294967291	3	49	0.30	776	0/10	0/10	0/10	0/10	128.0	1.5
32	4294967291	3	49	0.32	1209	0/10	0/10	0/10	0/10	103.0	1.5
32	4294967291	3	49	0.34	1884	0/10	0/10	0/10	0/10	90.8	1.3
32	4294967291	3	49	0.30 (negative control)	776	genuine 0/10 (raw candidates in 10)	–	–	–	126.7	1.5
40	4099511627689	1	9	0.04	3	10/10	10/10	10/10	10/10	6.9	2.1
40	4099511627689	1	9	0.06	5	10/10	10/10	10/10	10/10	4.7	2.1
40	4099511627689	1	9	0.08	9	10/10	0/10	0/10	10/10	3.7	1.4
40	4099511627689	1	9	0.10	15	2/10	0/10	0/10	2/10	3.2	1.4
40	4099511627689	1	9	0.12	27	1/10	0/10	0/10	2/10	2.5	1.4
40	4099511627689	1	9	0.14	48	0/10	0/10	0/10	0/10	1.5	0.8
40	4099511627689	1	9	0.16	84	0/10	0/10	0/10	0/10	1.4	0.6
40	4099511627689	1	9	0.18	147	0/10	0/10	0/10	0/10	1.2	0.6
40	4099511627689	1	9	0.20	255	0/10	0/10	0/10	0/10	1.0	0.6
40	4099511627689	1	9	0.22	445	0/10	0/10	0/10	0/10	0.9	0.7
40	4099511627689	1	9	0.24	776	0/10	0/10	0/10	0/10	0.9	0.7
40	4099511627689	1	9	0.26	1351	0/10	0/10	0/10	0/10	1.3	1.2

40	99511627689	1	9	0.28	2352	0/10	0/10	0/10	0/10	1.1	1.5
40	99511627689	1	9	0.30	4095	0/10	0/10	0/10	0/10	0.8	2.2
40	99511627689	1	9	0.32	7131	0/10	0/10	0/10	0/10	0.7	3.2
40	99511627689	1	9	0.34	12416	0/10	0/10	0/10	0/10	0.4	2.8
40	99511627689	1	9	0.30 (negative control)	4095	genuine 0/10 (raw candidates in 10)	–	–	–	0.6	1.6
40	99511627689	2	25	0.04	3	10/10	10/10	10/10	10/10	309.3	16.2
40	99511627689	2	25	0.06	5	10/10	10/10	10/10	10/10	230.1	16.3
40	99511627689	2	25	0.08	9	10/10	10/10	10/10	10/10	195.0	17.0
40	99511627689	2	25	0.10	15	10/10	10/10	10/10	10/10	175.8	18.0
40	99511627689	2	25	0.12	27	8/10	0/10	0/10	10/10	120.9	11.4
40	99511627689	2	25	0.14	48	1/10	0/10	0/10	3/10	74.0	11.8
40	99511627689	2	25	0.16	84	0/10	0/10	0/10	0/10	32.6	1.3
40	99511627689	2	25	0.18	147	0/10	0/10	0/10	0/10	28.7	0.7
40	99511627689	2	25	0.20	255	0/10	0/10	0/10	0/10	22.9	0.8
40	99511627689	2	25	0.22	445	0/10	0/10	0/10	0/10	21.3	0.8
40	99511627689	2	25	0.24	776	0/10	0/10	0/10	0/10	18.8	1.0
40	99511627689	2	25	0.26	1351	0/10	0/10	0/10	0/10	17.5	1.1
40	99511627689	2	25	0.28	2352	0/10	0/10	0/10	0/10	18.2	1.6
40	99511627689	2	25	0.30	4095	0/10	0/10	0/10	0/10	13.6	1.9
40	99511627689	2	25	0.32	7131	0/10	0/10	0/10	0/10	11.6	2.7
40	99511627689	2	25	0.34	12416	0/10	0/10	0/10	0/10	9.6	2.9
40	99511627689	2	25	0.30 (negative control)	4095	genuine 0/10 (raw candidates in 10)	–	–	–	13.3	1.8
40	99511627689	3	49	0.04	3	10/10	10/10	10/10	10/10	4980.1	99.2
40	99511627689	3	49	0.06	5	10/10	10/10	10/10	10/10	4301.2	91.3
40	99511627689	3	49	0.08	9	10/10	10/10	10/10	10/10	3432.4	91.0
40	99511627689	3	49	0.10	15	10/10	10/10	10/10	10/10	2799.4	92.2
40	99511627689	3	49	0.12	27	10/10	0/10	0/10	10/10	1854.1	85.9
40	99511627689	3	49	0.14	48	5/10	0/10	0/10	10/10	1043.5	57.8
40	99511627689	3	49	0.16	84	0/10	0/10	0/10	0/10	347.8	1.7
40	99511627689	3	49	0.18	147	0/10	0/10	0/10	0/10	269.7	1.3
40	99511627689	3	49	0.20	255	0/10	0/10	0/10	0/10	227.2	1.3
40	99511627689	3	49	0.22	445	0/10	0/10	0/10	0/10	202.1	1.4
40	99511627689	3	49	0.24	776	0/10	0/10	0/10	0/10	182.1	1.4
40	99511627689	3	49	0.26	1351	0/10	0/10	0/10	0/10	164.1	1.6
40	99511627689	3	49	0.28	2352	0/10	0/10	0/10	0/10	156.1	2.2
40	99511627689	3	49	0.30	4095	0/10	0/10	0/10	0/10	143.2	2.6
40	99511627689	3	49	0.32	7131	0/10	0/10	0/10	0/10	132.2	3.7
40	99511627689	3	49	0.34	12416	0/10	0/10	0/10	0/10	109.4	4.1

4099511627689	3	49	0.30 (negative control)	4095	genuine 0/10 (raw candidates in 10)	–	–	–	137.8	2.6
7205759403792793	1	9	0.04	4	10/10	10/10	10/10	10/10	5.9	1.9
7205759403792793	1	9	0.06	10	10/10	10/10	10/10	10/10	4.6	1.8
7205759403792793	1	9	0.08	22	10/10	0/10	0/10	10/10	4.6	1.7
7205759403792793	1	9	0.10	48	0/10	0/10	0/10	0/10	3.5	1.3
7205759403792793	1	9	0.12	105	0/10	0/10	0/10	0/10	3.1	1.4
7205759403792793	1	9	0.14	229	0/10	0/10	0/10	0/10	2.2	0.9
7205759403792793	1	9	0.16	497	0/10	0/10	0/10	0/10	2.0	0.7
7205759403792793	1	9	0.18	1082	0/10	0/10	0/10	0/10	1.7	0.8
7205759403792793	1	9	0.20	2352	0/10	0/10	0/10	0/10	1.7	1.2
7205759403792793	1	9	0.22	5113	0/10	0/10	0/10	0/10	1.4	2.0
7205759403792793	1	9	0.24	11113	0/10	0/10	0/10	0/10	1.7	4.0
7205759403792793	1	9	0.26	24154	0/10	0/10	0/10	0/10	1.1	7.5
7205759403792793	1	9	0.28	52498	0/10	0/10	0/10	0/10	0.9	14.6
7205759403792793	1	9	0.30	114104	0/10	0/10	0/10	0/10	0.8	30.1
7205759403792793	1	9	0.32	248003	0/10	0/10	0/10	0/10	0.8	69.8
7205759403792793	1	9	0.34	539027	0/10	0/10	0/10	0/10	0.5	97.0
7205759403792793	1	9	0.30 (negative control)	114104	genuine 0/10 (raw candidates in 10)	–	–	–	0.9	30.4
7205759403792793	2	25	0.04	4	10/10	10/10	10/10	10/10	412.0	17.2
7205759403792793	2	25	0.06	10	10/10	10/10	10/10	10/10	345.4	16.8
7205759403792793	2	25	0.08	22	10/10	10/10	10/10	10/10	288.2	16.6
7205759403792793	2	25	0.10	48	10/10	10/10	10/10	10/10	227.0	17.4
7205759403792793	2	25	0.12	105	6/10	0/10	0/10	9/10	158.4	11.4
7205759403792793	2	25	0.14	229	0/10	0/10	0/10	1/10	95.9	11.7
7205759403792793	2	25	0.16	497	0/10	0/10	0/10	0/10	47.0	1.6
7205759403792793	2	25	0.18	1082	0/10	0/10	0/10	0/10	40.5	1.1
7205759403792793	2	25	0.20	2352	0/10	0/10	0/10	0/10	31.9	1.5
7205759403792793	2	25	0.22	5113	0/10	0/10	0/10	0/10	26.5	2.2
7205759403792793	2	25	0.24	11113	0/10	0/10	0/10	0/10	24.3	4.1
7205759403792793	2	25	0.26	24154	0/10	0/10	0/10	0/10	24.5	8.6
7205759403792793	2	25	0.28	52498	0/10	0/10	0/10	0/10	21.1	16.4
7205759403792793	2	25	0.30	114104	0/10	0/10	0/10	0/10	18.4	34.0
7205759403792793	2	25	0.32	248003	0/10	0/10	0/10	0/10	15.3	73.8
7205759403792793	2	25	0.34	539027	0/10	0/10	0/10	0/10	12.2	98.6
7205759403792793	2	25	0.30 (negative control)	114104	genuine 0/10 (raw candidates in 10)	–	–	–	21.9	38.5
7205759403792793	3	49	0.04	4	10/10	10/10	10/10	10/10	7418.9	87.1
7205759403792793	3	49	0.06	10	10/10	10/10	10/10	10/10	6033.7	90.1

72057594037927931	3	49	0.08	22	10/10	10/10	10/10	10/10	4967.4	99.0
72057594037927931	3	49	0.10	48	10/10	10/10	10/10	10/10	3987.9	89.9
72057594037927931	3	49	0.12	105	10/10	6/10	3/10	10/10	2857.7	93.2
72057594037927931	3	49	0.14	229	2/10	0/10	0/10	8/10	1600.4	66.6
72057594037927931	3	49	0.16	497	0/10	0/10	0/10	0/10	475.2	2.0
72057594037927931	3	49	0.18	1082	0/10	0/10	0/10	0/10	390.3	1.7
72057594037927931	3	49	0.20	2352	0/10	0/10	0/10	0/10	285.0	2.0
72057594037927931	3	49	0.22	5113	0/10	0/10	0/10	0/10	251.5	2.9
72057594037927931	3	49	0.24	11113	0/10	0/10	0/10	0/10	233.3	4.9
72057594037927931	3	49	0.26	24154	0/10	0/10	0/10	0/10	235.1	9.9
72057594037927931	3	49	0.28	52498	0/10	0/10	0/10	0/10	209.6	20.9
72057594037927931	3	49	0.30	114104	0/10	0/10	0/10	0/10	204.2	42.5
72057594037927931	3	49	0.32	248003	0/10	0/10	0/10	0/10	183.4	96.8
72057594037927931	3	49	0.34	539027	0/10	0/10	0/10	0/10	152.6	151.3
72057594037927931	3	49	0.30 (negative control)	114104	genuine 0/10 (raw candidates in 10)	–	–	–	191.9	42.1

Observations (no interpretation)

- Every planted cell with β ≥ 0.20 recorded 0/10 on every metric, at every size and multiplicity (96 cells, 960 trials). The largest β with any recovery is 0.18 (1/10, 24-bit, $\ell = 2$).
- Largest β with `recovered ≥ 5/10`, by size and ℓ: 24-bit 0.12 / 0.14 / 0.14; 32-bit 0.08 / 0.12 / 0.14; 40-bit 0.08 / 0.12 / 0.14; 56-bit 0.08 / 0.12 / 0.12 ($\ell = 1$ / 2 / 3).
- Largest β with `HG first two = 10/10`: $\ell = 1$: 0.06 at every size; $\ell = 2$: 0.08 (24, 32 bits), 0.10 (40, 56 bits); $\ell = 3$: 0.10 at every size.
- Negative control: 0 genuine pairs in 120 trials; every trial reported the raw candidate (0, 0), which fails $S_3 \equiv 0 \pmod p$ and is filtered.
- Mean LLL time per trial falls with β and rises with the prime size; the largest cell mean is 7.4 s (56-bit, $\ell = 3$, β = 0.04). Extraction time grows with B at 56 bits (up to 151 ms at B = 539027).